

```
from google.colab import files
uploaded = files.upload()
```

Choose Files 4 files

```
olist_customers_dataset.csv(text/csv) - 9033957 bytes, last modified: 10/1/2021 - 100% done
olist_order_reviews_dataset.csv(text/csv) - 14451670 bytes, last modified: 10/1/2021 - 100% done
olist_orders_dataset.csv(text/csv) - 17654914 bytes, last modified: 10/1/2021 - 100% done
olist_products_dataset.csv(text/csv) - 2379446 bytes, last modified: 10/1/2021 - 100% done
Saving olist_customers_dataset.csv to olist_customers_dataset (1).csv
Saving olist_order_reviews_dataset.csv to olist_order_reviews_dataset.csv
Saving olist_orders_dataset.csv to olist_orders_dataset.csv
Saving olist_products_dataset.csv to olist_products_dataset.csv
```

```
import pandas as pd
```

```
orders = pd.read_csv('olist_orders_dataset.csv')
reviews = pd.read_csv('olist_order_reviews_dataset.csv')
customers = pd.read_csv('olist_customers_dataset.csv')

master_df = orders.merge(reviews, on='order_id', how='left')
master_df = master_df.merge(customers, on='customer_id', how='left')

print(f"Master Dataset Shape: {master_df.shape}")
print(f"Duplicate Order IDs: {master_df['order_id'].duplicated().sum()}")
```

```
Master Dataset Shape: (99992, 18)
Duplicate Order IDs: 551
```

```
master_df = master_df.drop_duplicates(subset=['order_id'])

date_columns = ['order_estimated_delivery_date', 'order_delivered_customer_date']
for col in date_columns:
    master_df[col] = pd.to_datetime(master_df[col])

master_df = master_df.dropna(subset=['order_delivered_customer_date'])

master_df['Days_Difference'] = (master_df['order_estimated_delivery_date'] - master_df['order_delivered_customer_date']).dt.days

def classify_status(days):
    if days >= 0:
        return 'On Time'
    elif days > -5:
        return 'Late'
    else:
        return 'Super Late'

master_df['Delivery_Status'] = master_df['Days_Difference'].apply(classify_status)

print(master_df['Delivery_Status'].value_counts())
```

```
Delivery_Status
On Time      88649
Super Late   4665
Late         3162
Name: count, dtype: int64
```

```
state_performance = master_df.groupby('customer_state')['Delivery_Status'].value_counts(normalize=True).unstack().fillna(0)
state_performance['Late_Total %'] = (state_performance.get('Late', 0) + state_performance.get('Super Late', 0)) * 100
state_performance_sorted = state_performance.sort_values(by='Late_Total %', ascending=False)

print(state_performance_sorted[['Late_Total %']].head(10))
```

```
Delivery_Status  Late_Total %
customer_state
AL               23.929471
MA               19.665272
PI               15.966387
CE               15.324472
SE               15.223881
BA               14.035627
RJ               13.470412
TO               12.773723
PA               12.367865
ES               12.230576
```

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```
from google.colab import files
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```

Choose Files 2 files

olist_order_items_dataset.csv(text/csv) - 15438671 bytes, last modified: 10/1/2021 - 100% done
product_category_name_translation.csv(text/csv) - 2613 bytes, last modified: 4/24/2026 - 100% done
Saving olist_order_items_dataset.csv to olist_order_items_dataset.csv
Saving product_category_name_translation.csv to product_category_name_translation.csv

```
import pandas as pd

if 'review_score' not in master_df.columns:
    reviews = pd.read_csv('olist_order_reviews_dataset.csv')
    master_df = master_df.merge(reviews[['order_id', 'review_score']], on='order_id', how='left')

sentiment_analysis = master_df.groupby('Delivery_Status')['review_score'].mean()

print("Average Customer Rating by Delivery Performance:")
print(sentiment_analysis)
```

Average Customer Rating by Delivery Performance:	
Delivery_Status	
Late	3.593231
On Time	4.294358
Super Late	1.854967
Name: review_score, dtype: float64	

```
import pandas as pd

products = pd.read_csv('olist_products_dataset.csv')
translation = pd.read_csv('product_category_name_translation.csv')
order_items = pd.read_csv('olist_order_items_dataset.csv')

products_en = products.merge(translation, on='product_category_name', how='left')

master_df = master_df.drop(columns=['product_id', 'product_category_name_english'], errors='ignore')
master_df = master_df.merge(order_items[['order_id', 'product_id']], on='order_id', how='left')
master_df = master_df.merge(products_en[['product_id', 'product_category_name_english']], on='product_id', how='left')

cat_perf = master_df.groupby('product_category_name_english')['Delivery_Status'].value_counts(normalize=True).unstack().fillna(0)
cat_perf['Late_Total %'] = (cat_perf.get('Late', 0) + cat_perf.get('Super Late', 0)) * 100

print(cat_perf.sort_values(by='Late_Total %', ascending=False).head(10))
```

Delivery_Status	Late	On Time	Super Late	\
product_category_name_english				
home_comfort_2	0.033333	0.833333	0.133333	
furniture_mattress_and_upholstery	0.027027	0.864865	0.108108	
audio	0.035912	0.872928	0.091160	
fashion_underwear_beach	0.031496	0.874016	0.094488	
christmas_supplies	0.040000	0.880000	0.080000	
books_technical	0.060837	0.889734	0.049430	
home_comfort	0.030303	0.897436	0.072261	
construction_tools_lights	0.063123	0.900332	0.036545	
food	0.066132	0.901804	0.032064	

electronics	0.046904	0.902528	0.050568
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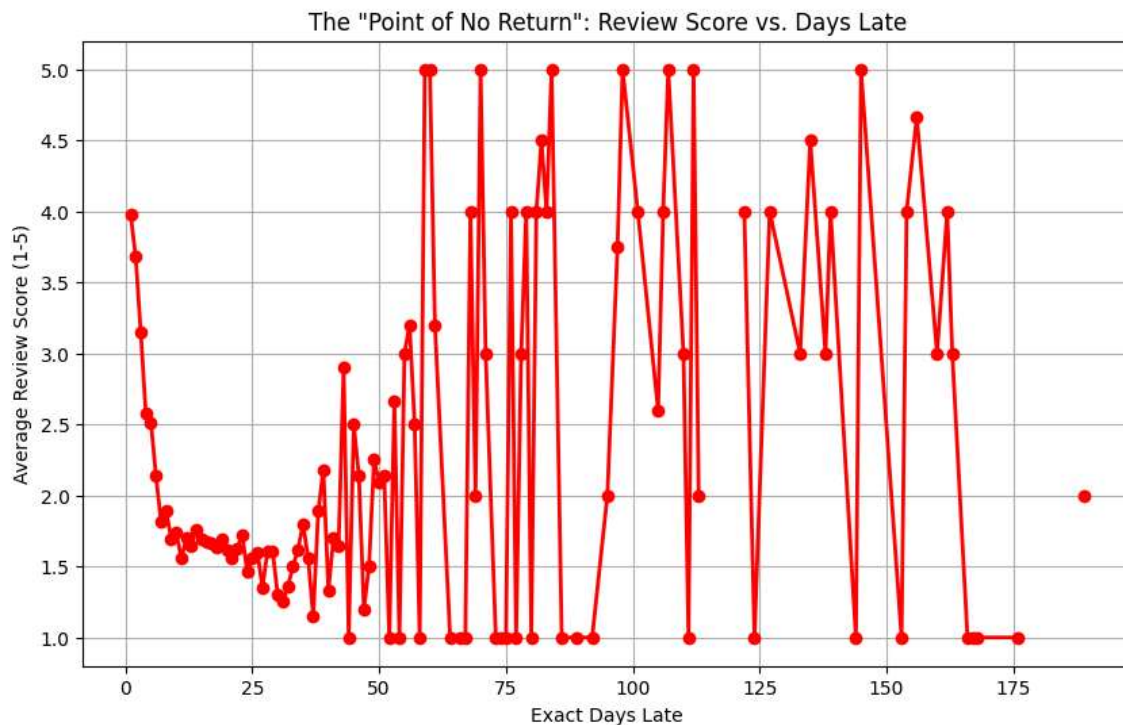
Delivery_Status	Late_Total %
product_category_name_english	
home_comfort_2	16.666667
furniture_mattress_and_upholstery	13.513514
audio	12.707182
fashion_underwear_beach	12.598425
christmas_supplies	12.000000
books_technical	11.026616
home_comfort	10.256410
construction_tools_lights	9.966777
food	9.819639
electronics	9.747160

```
import matplotlib.pyplot as plt
```

```
late_orders = master_df[master_df['Days_Difference'] < 0].copy()
late_orders['Days_Late'] = late_orders['Days_Difference'].abs()
```

```
decay_data = late_orders.groupby('Days_Late')['review_score'].mean().reset_index()
```

```
plt.figure(figsize=(10, 6))
plt.plot(decay_data['Days_Late'], decay_data['review_score'], marker='o', color='red', linewidth=2)
plt.title('The "Point of No Return": Review Score vs. Days Late')
plt.xlabel('Exact Days Late')
plt.ylabel('Average Review Score (1-5)')
plt.grid(True)
plt.show()
```



```
master_df.to_csv('veridi_logistics_audit_final.csv', index=False)
```

```
from google.colab import files
files.download('veridi_logistics_audit_final.csv')
```

```
print("Success! Your final audit data is downloading to your laptop.")
```

Success! Your final audit data is downloading to your laptop.

